Stargazing Field Guide: Finding Your Way Around the Sky from India

What you will be able to do

1. Plan a successful observing session (Moon, weather, site, light pollution).
2. Orient yourself and find north **without a compass**.
3. **Star-hop** from bright signposts to fainter constellations in every season.
4. Use an **all-sky chart, planisphere or app** correctly.
5. Keep a proper **observing log**.

1. Before you go out: planning checklist

Check	Why
Moon phase	A bright Moon washes out faint stars. Best nights: Krishna Ashtami to Shukla Panchami (around Amavasya). The Moon itself is best viewed from Shukla Chaturthi to Ekadashi.
Weather	Check cloud cover (IMD, Windy, Clear Outside). **October–March is the best season in most of India**; the monsoon often clouds out the sky.
Site	Away from street lights, with an open view of the horizon you need. A rooftop works in towns.
Light pollution	Use `https://www.lightpollutionmap.info` to find darker places nearby.
Charts / app	Stellarium (free), Sky Map, SkySafari, Star Walk — set your location; download offline.
Red torch	Cover a torch with red cellophane, or use your app's red "night mode".
Comfort & safety	Mat or reclining chair, warm clothes, mosquito repellent, water; go in a group and tell someone where you are.

How dark is your sky? A quick test

Look at the **Little Dipper** after 30 minutes of dark adaptation:

What you can see	Sky quality
Only Polaris and Kochab	city sky — faintest stars about magnitude 3
All 4 bowl stars	suburban — about 4.5
All 7 stars and a faint Milky Way	rural — about 5.5–6
Milky Way full of structure; the Andromeda Galaxy obvious	truly dark (Ladakh, Spiti, Rann of Kutch)

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P.T.O.

India's dark-sky treasure

The **Hanle Dark Sky Reserve** in Ladakh (declared in 2022) is India's first dark-sky reserve, around the Indian Astronomical Observatory at 4,500 m.

2. Your eyes: dark adaptation

- Pupils widen within seconds, but the eye's chemical sensitivity takes **20–30 minutes** to build up fully.
- **One glance at a white screen or torch resets it.** Use red light only.
- **Averted vision:** look slightly to one side of a faint object — the edge of the retina is more sensitive to dim light.

3. Finding directions without a compass

1. **Sunset point**: roughly west (north-west in summer, south-west in winter).
2. **Polaris**: true north within 1° (Saptarishi pointers or the "W" — Lecture 1.1).
3. **Orion's Belt**: Mintaka lies almost on the celestial equator, so it rises **due east** and sets **due west** from anywhere on Earth.
4. **The Southern Cross** (April–June, from south of $\sim 25^\circ$ N): its long axis points towards the south celestial pole.

4. Star-hopping routes (season by season, ~9 PM)

Star-hopping means moving from a star you know to the next, using shapes, lines and hand-measured distances. Learn one route per season.

Winter (December–February): start from ORION

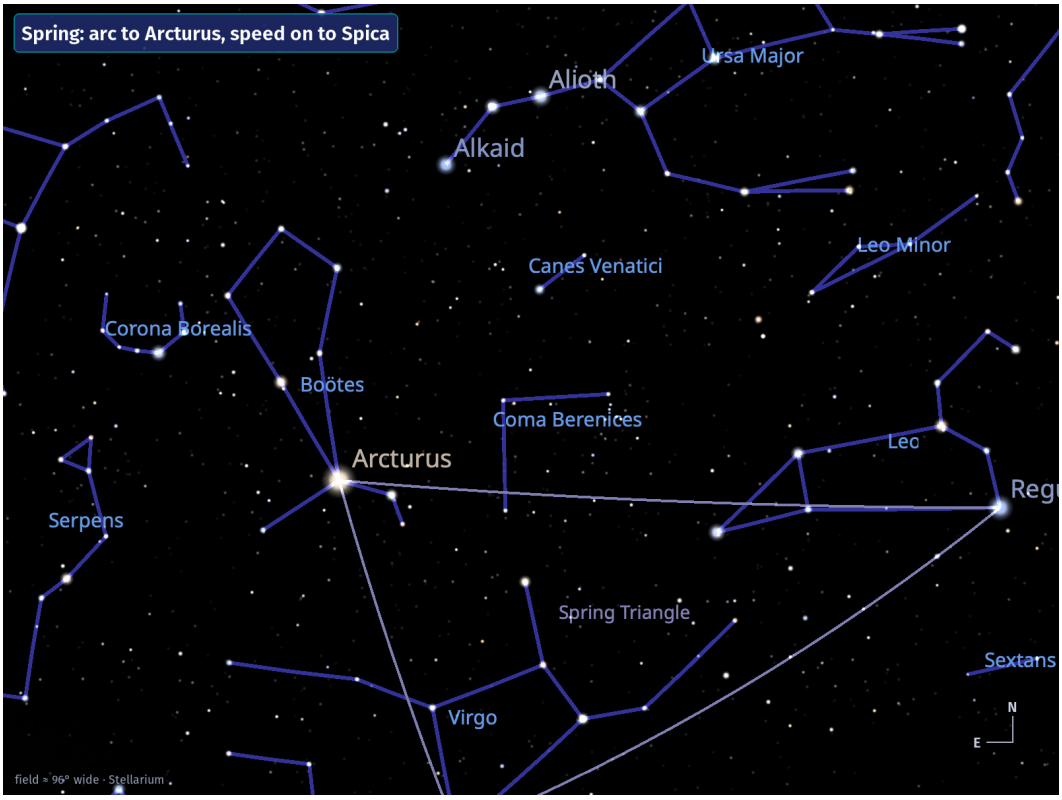
- **Belt → down-left,** $\sim 20^\circ$ → **Sirius**, the brightest star.
- **Belt → up-right,** $\sim 20^\circ$ → **Aldebaran**, then on $\sim 15^\circ$ to the **Pleiades**.
- **Rigel → Betelgeuse**, and the same distance again (bending slightly east) → **Castor & Pollux**.
- **Betelgeuse → east** $\sim 25^\circ$ → **Procyon**, completing the **Winter Triangle**.
- **Middle of the Belt → straight north through Orion's head,** $\sim 35^\circ$ **more** → yellow **Capella**, nearly overhead.
- Low in the south, look for brilliant **Canopus**.

Spring (March–May): start from the BIG DIPPER

- Continue the curve of the handle $\sim 30^\circ$ → orange **Arcturus**; keep going $\sim 30^\circ$ → blue-white **Spica**.
- Pointers *backwards* (Dubhe → Merak and on, $\sim 45^\circ$) → the **Sickle of Leo** and **Regulus**.
- **Spring Triangle**: Arcturus–Spica–Regulus (or Denebola). **Corvus** is a small quadrilateral $\sim 15^\circ$ west of Spica.
- Late spring, low in the south: the **Southern Cross** and its two **Pointers**.

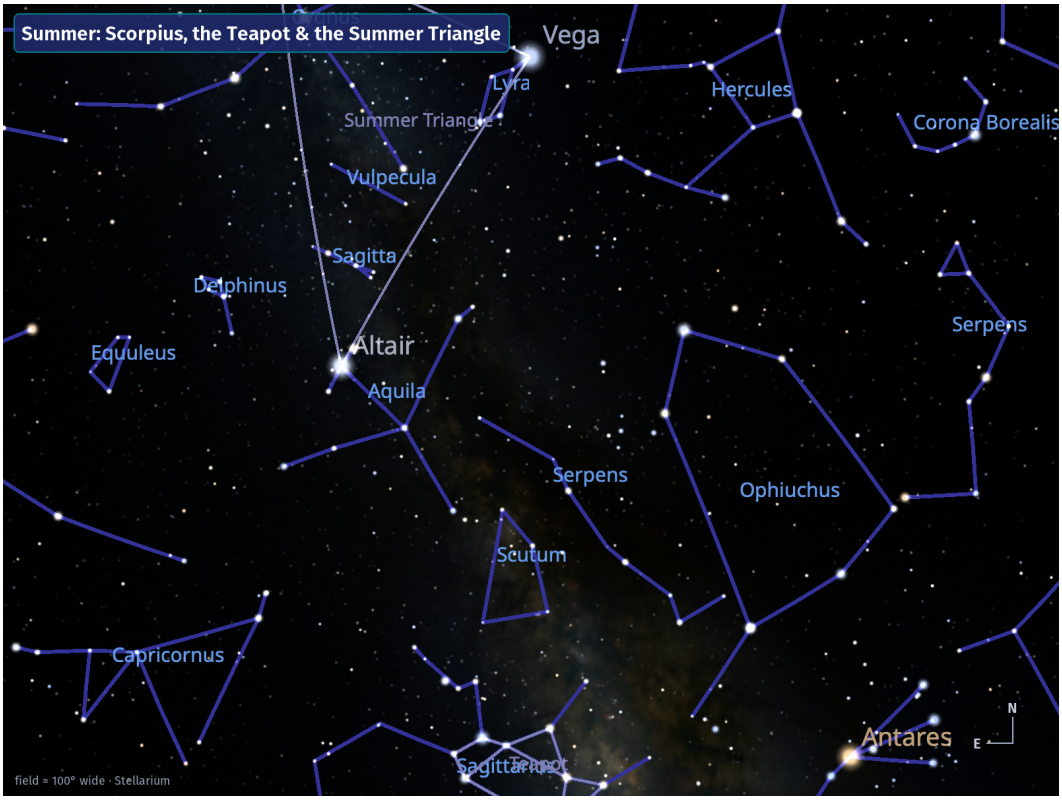


Winter hops from Orion (Stellarium).



“Follow the arc to Arcturus, then speed on to Spica” (Stellarium).

Summer / monsoon (June–August): SCORPIUS and the SUMMER TRIANGLE



Scorpius, the Teapot and the Summer Triangle (Stellarium).

- Low in the south: the curved tail of **Scorpius** with red **Antares**. East (left) of the sting: the **Teapot** of Sagittarius, with the Milky Way rising like steam from the spout.
- High in the east: the **Summer Triangle** — Vega (brightest), Altair (with a fainter star on each side), Deneb (top of the Northern Cross).
- **Vega → west** ~ 20° → the four-star **Keystone of Hercules** (it contains the globular cluster M13).

Autumn (September–November): the GREAT SQUARE

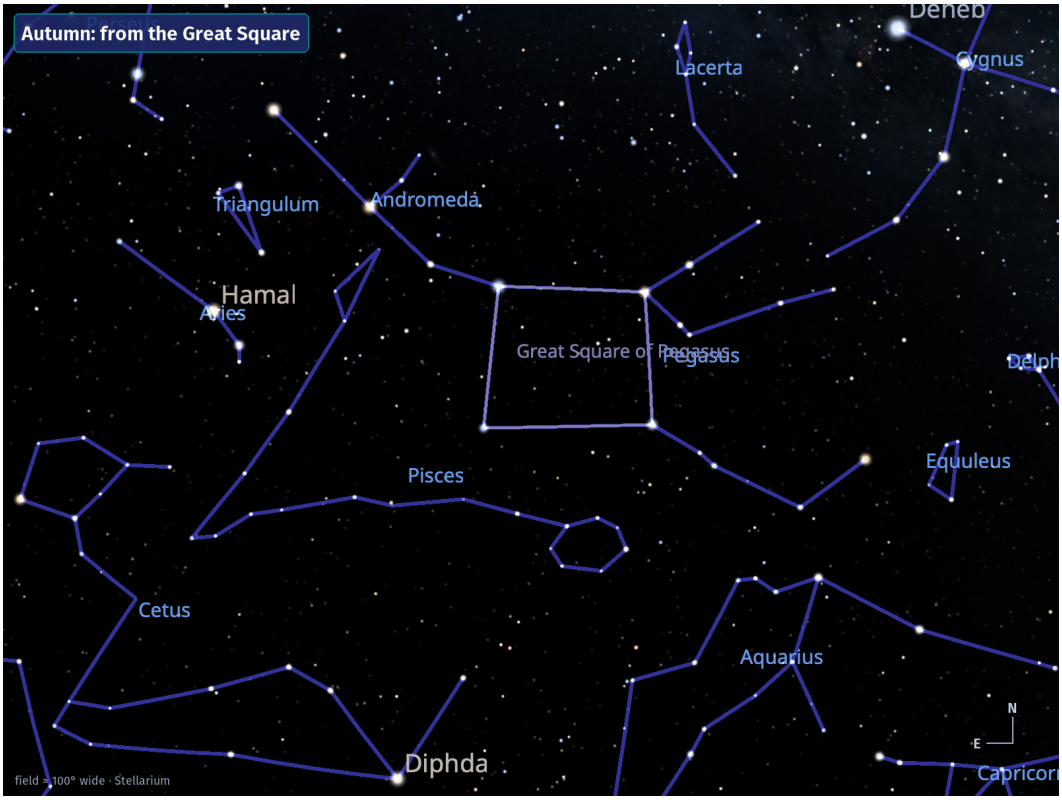
- High in the east: a large, almost empty **square** of four stars.
- From **Alpheratz** (north-east corner) follow the chain east to **Mirach**, then turn north two stars → the **Andromeda Galaxy**.
- From the square's west side (Scheat → Markab), go south ~ 45° → lonely **Fomalhaut**.
- **Cassiopeia's W** is low in the north; the **Summer Triangle** is still in the west.

All year: finding Polaris

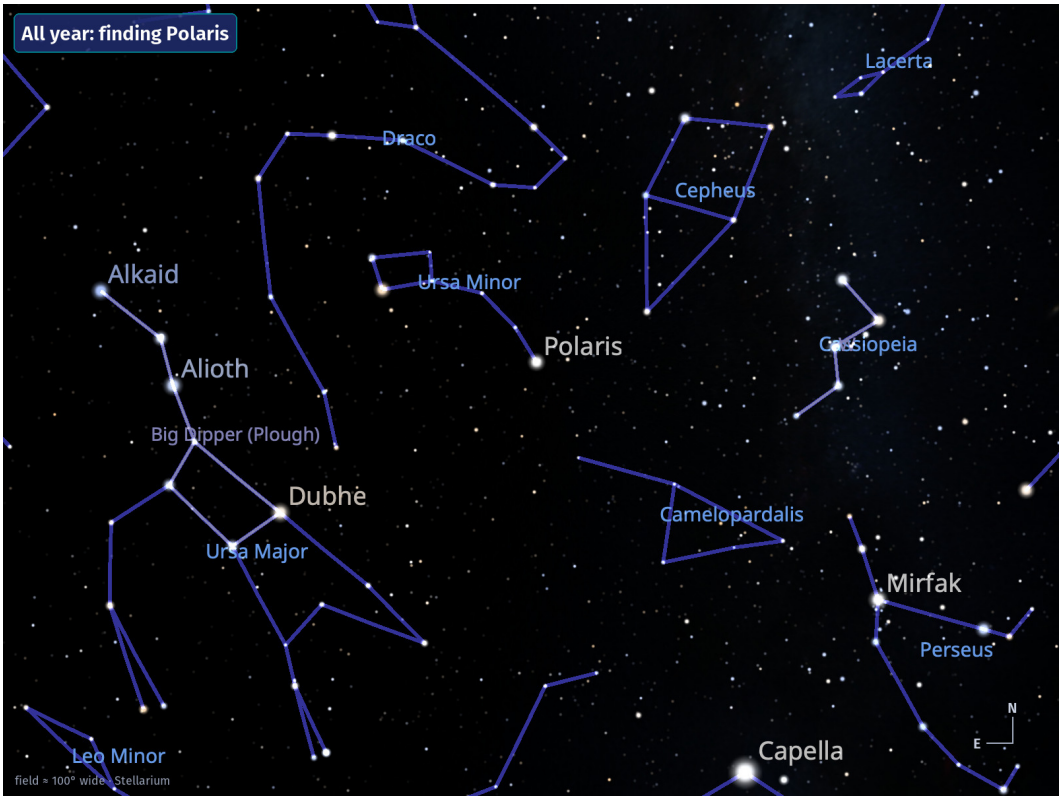
5. Using all-sky charts, planispheres and apps

These four charts show the sky from India (latitude 20° N) at **9 PM IST** in each season:

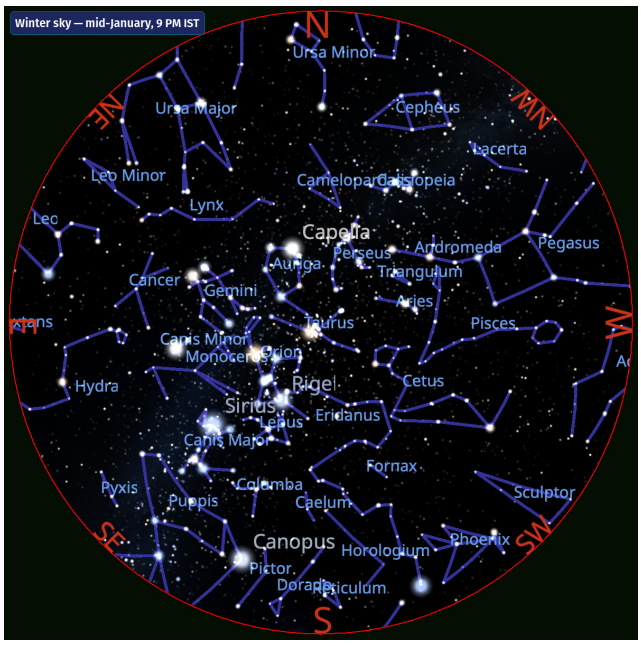
- **How to hold an all-sky chart:** face a direction, e.g. South; turn the chart so "S" is at the **bottom**; hold it up against the sky. **East is on the left** because you are looking *up*, unlike a ground map.
- **Each chart works for other dates too:** the same sky appears **2 hours earlier each month later** (Lecture 1.1). The January 9 PM chart also shows 11 PM in mid-December and 7 PM in mid-February.
- **Apps:** turn on night (red) mode, switch off auto-brightness, and use them to *check* what you see — don't stare at the screen all night.



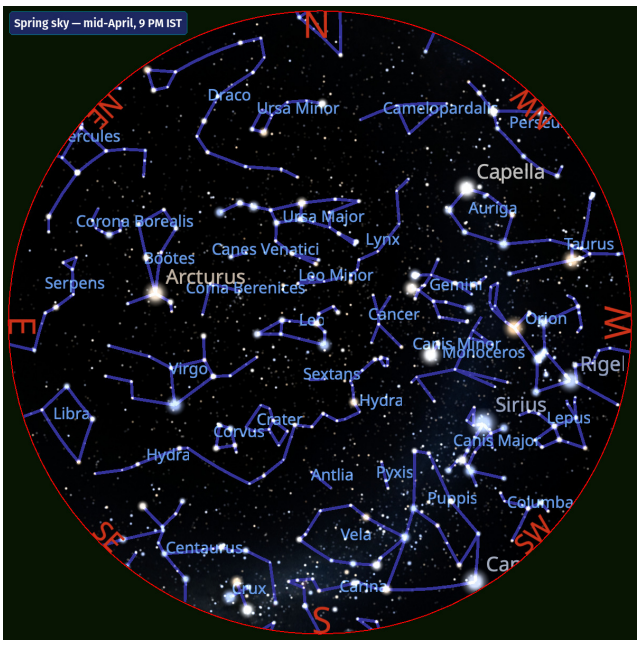
Autumn hops from the Great Square of Pegasus (Stellarium).



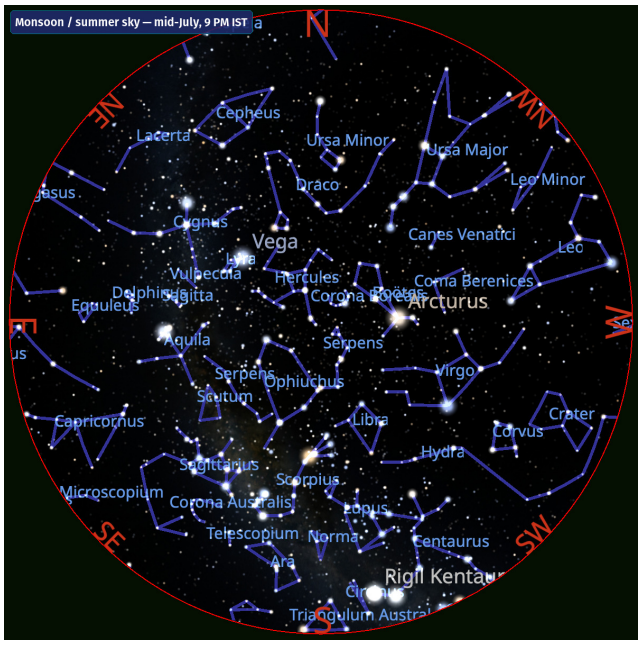
The Big Dipper and Cassiopeia lie on opposite sides of Polaris (Stellarium).



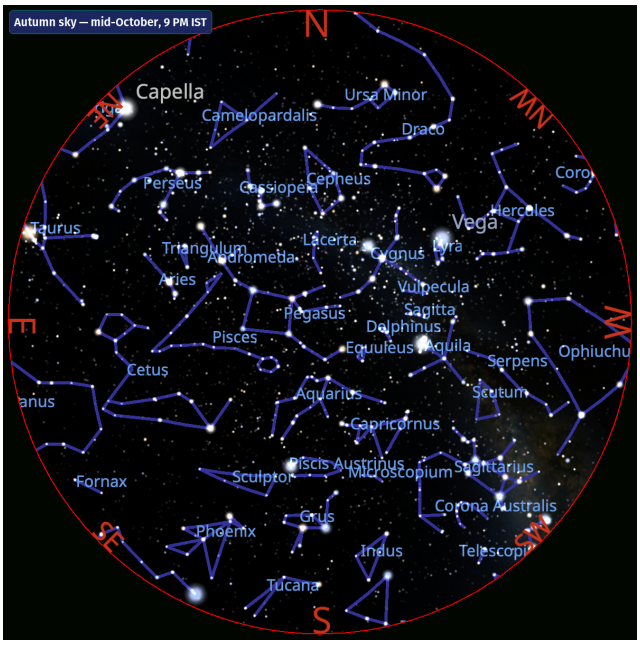
Winter (mid-January)



Spring (mid-April)



Summer / monsoon (mid-July)



Autumn (mid-October)

Seasonal all-sky charts made with Stellarium. Centre = overhead (zenith); edge = horizon.

6. What to look for during the course

Planets

Period	Evening sky (after sunset)	Morning sky (before sunrise)
Oct–Dec 2026	**Saturn** (Cetus/Pisces, high in the south-east)	Jupiter and Mars in Leo/Cancer; Venus from November
Jan 2027	Saturn in the west	**Venus** brilliant in the south-east
Feb–Mar 2027	**Jupiter AND Mars at opposition in Leo** — up all night near Regulus	Venus (morning star)
Apr–Jun 2027	Mars and Jupiter in Leo, sinking into the west	Saturn returns (Pisces)

(continued)

Period	Evening sky (after sunset)	Morning sky (before sunrise)
Jul–Sep 2027	Mars (Virgo/Libra) in the west	Saturn high before dawn

"Opposition" means a planet is opposite the Sun: it rises at sunset, is up all night and is at its closest and brightest. (Computed for the 15th of each month.)

Meteor showers (best after midnight, away from city lights)

Shower	Peak	Rate (dark sky)	Notes
Orionids	~21 Oct	~20/h	debris of Halley's Comet
Leonids	~17 Nov	~15/h	
Geminids	**13–14 Dec**	up to ~120/h	best of the year; in 2026 the crescent Moon sets early — excellent!
Quadrantids	3–4 Jan	~80/h	short peak; radiant low from India
Lyrids	~22 Apr	~18/h	
Eta Aquariids	5–6 May	~40/h	favours tropical latitudes — good from India
Perseids	12–13 Aug	~100/h	monsoon clouds are the main enemy

How to watch: lie flat, look about halfway up the sky, allow 30 minutes for dark adaptation, and count meteors in 15-minute blocks. No telescope is needed!

The International Space Station

The ISS looks like a very bright, steady, non-blinking point crossing the sky in 3–6 minutes, usually just after dusk or before dawn. Find pass times with NASA's "Spot the Station" or `https://www.heavens-above.com`.

7. The observing log

A log turns stargazing into **data**. Always record: date, time (IST), place (with latitude/longitude), sky conditions (clouds, Moon, faintest star seen), equipment, what you saw, a sketch, and your notes and questions. A printable log sheet is provided with the course resources.

8. Etiquette at a star party

- **No white lights** — dim red only. Car headlights are a crime!
- Ask before touching anyone's telescope; never bump the tripod.
- Laser pointers only by instructors, briefly, never towards aircraft or people.
- Leave the site clean — and **share**: show the Moon or Saturn to someone who has never seen it.

Key points to remember

- Plan around the Moon and the weather; **October–March** is the prime season in India.
- Dark-adapt for 20–30 minutes; red light; averted vision.
- One signpost per season: Orion (winter), Big Dipper (spring), Scorpius / Summer Triangle (summer), Great Square (autumn).

Key points to remember *(continued)*

- *Arc to Arcturus, speed on to Spica*; Merak → Dubhe ×5 → Polaris; Orion's Belt → Sirius / Aldebaran.
- Feb–Mar 2027: Mars and Jupiter at opposition in Leo. Geminids: 13–14 Dec 2026.

Further reading (free / public)

- **Stellarium** User Guide (free) — `https://stellarium.org`
- International Meteor Organization calendar — `https://www.imo.net`
- T. Dickinson, *NightWatch: A Practical Guide to Viewing the Universe* (Firefly Books); R. W. Sinnott, *Sky & Telescope's Pocket Sky Atlas*.